

# MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

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ABSTRACT. The exact law state is augmented by the cumulative actual-contact ledger. The BBGKY perturbed test solves a terminal-value Poisson equation with the backward observable propagator constructed by forward evolution in remaining time; it has zero terminal remainder and never invokes a negative-time hierarchy. Normal convergence is proved in an absolute generator graph norm. Analytic cylinders form the core and smooth cylinders are approximated on action compacta. A bounded-gradient Tataru penalty keeps every doubled cotangent inside a uniform exponential Hamiltonian chart, yielding comparison and nonlinear semigroup convergence.

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## 1. ROUND-SEVEN RECONSTRUCTION: A FORWARD TERMINAL-VALUE CORRECTOR, AN AUGMENTED LAW STATE, AND BOUNDED-COTANGENT COMPARISON

This section replaces the round-six controlling module. The corrector solves a time-dependent terminal-value Poisson equation and uses only the

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forward hierarchy semigroup. The exact state carries the cumulative contact ledger. The generator theorem is first proved on an analytic law-coordinate algebra; nonanalytic cylinders are reached by graph-norm approximation on compact containment sets. Comparison uses a bounded-gradient Tataru penalty, so the collision increment never leaves the exponential Hamiltonian core.

**1.1. The augmented exact law state.** Let

$$\widehat{Z}_t^\varepsilon = (Z_t^\varepsilon, A_t^\varepsilon)$$

where  $Z_t^\varepsilon$  is the current hard-sphere microstate and  $A_t^\varepsilon$  is the cumulative marked actual-contact measure up to time  $t$ . The deterministic flow on the augmented state is Markov and invertible between impacts. Its exact law is  $\widehat{\rho}_t^\varepsilon$ . The hierarchy coordinates consist of all current factorial correlations together with all mixed correlations having one or more ledger marks. Thus a density/contact path cylinder is a function of the present augmented law, not an undeclared past functional.

Let  $\mathfrak{A}_{\alpha,\beta}^{\text{an}}$  be the completion of finite polynomials in those coordinates under the absolute coefficient graph norm

$$\|F\|_{\text{gr}} = \sum_J |c_J| w_J + \sum_J |c_J| \|\mathbb{A}_\varepsilon M_J\| w_J,$$

where  $\mathbb{A}_\varepsilon$  is the full augmented BBGKY generator. The weights include the factorial label radius and the B2 measure-trace norm.

**Lemma 1.1** (Analytic law algebra). *The space  $\mathfrak{A}_{\alpha,\beta}^{\text{an}}$  is a Banach algebra, is stable under exponentiation of its bounded elements, and is contained in the graph domain of  $\mathbb{A}_\varepsilon$ . The generator acts termwise and normally on compact measure-trace balls. No generic  $C_b^3$  cylinder is declared to be analytic merely because it is smooth.*

*Proof.* The product of two hierarchy monomials is their shuffle product. Factorial weights and the binomial identity bound the shuffle coefficient by the product of the two absolute coefficient norms. The B2 generator raises label and ledger degree by at most one, and its graph image is already included in the weight. Hence products and generator images converge absolutely. The exponential series of a bounded algebra element is normally convergent in the same graph norm.  $\square$

**1.2. The forward terminal-value corrector.** Let  $A(t)$  be the limiting triangular hierarchy generator on observables. For  $t \leq s$ , let  $V(t, s)$  be the backward observable propagator satisfying

$$\partial_t V(t, s) = -A(t)V(t, s), \quad V(s, s) = I.$$

It is constructed by a forward Picard iteration in the remaining-time variable  $r = s - t$  and does not require a negative-time hierarchy evolution. If the

uncancelled order- $j$  defect is  $D_j(t)$ , define

$$c_j(t) = \int_t^T V(t, s) D_j(s) ds, \quad c_j(T) = 0.$$

Then

$$\partial_t c_j(t) + A(t) c_j(t) = -D_j(t).$$

The terminal boundary term is exactly zero.

**Theorem 1.2** (Full triangular forward corrector). *For every bounded analytic cylinder  $F$  and every finite horizon, the triangular family  $c_j$  is normally summable in the absolute graph norm and the perturbed test*

$$F_\varepsilon(t) = F(t) + \sum_{j \geq 1} \mu_\varepsilon^{-j} c_j(t)$$

*belongs to the full augmented generator domain. Its logarithmic generator satisfies*

$$\mathcal{H}_\varepsilon F_\varepsilon = \mathcal{H}F + o(1)$$

*locally uniformly on measure-trace compact sets. No negative-time BBGKY operator and no finite-time pseudo-inverse are used.*

*Proof.* Differentiate the Duhamel integral. The lower endpoint gives  $-D_j(t)$  and  $\partial_t V(t, s) = -A(t)V(t, s)$  gives

$$\partial_t c_j(t) = -D_j(t) - A(t) c_j(t).$$

Thus the defect is cancelled with no terminal remainder. The forward remaining-time construction gives

$$\|c_j(t)\|_{\text{gr}} \leq \int_t^T e^{C(s-t)} \|D_j(s)\|_{\text{gr}} ds.$$

Connected coefficient bounds yield  $\|D_j\|_{\text{gr}} \leq C\rho^j$  with  $\rho < \mu_\varepsilon$  on the chosen label ball, so the series of both  $c_j$  and  $\mathbb{A}_\varepsilon c_j$  converges. Substitution in the logarithmic generator cancels every triangular defect at the same order. The remaining B2 consistency and cyclic errors tend to zero.  $\square$

**1.3. Approximation of nonanalytic cylinders.** Let  $K_M$  be a compact containment set in the weak law topology with bounded energy and action. A  $C_b^3$  cylinder depends on finitely many bounded coordinates in a compact box. Multivariate Bernstein polynomials approximate it together with first and second derivatives on that box. Multiply by an entire cutoff equal to one on the coordinate projection of  $K_M$ .

**Lemma 1.3** (Graph-core approximation). *For every finite-coordinate  $C_b^3$  cylinder  $F$  there are  $F^{(n)} \in \mathfrak{A}_{\alpha, \beta}^{\text{an}}$  such that, on  $K_M$ ,*

$$F^{(n)} \rightarrow F, \quad DF^{(n)} \rightarrow DF, \quad D^2 F^{(n)} \rightarrow D^2 F,$$

*and their collision increments are bounded uniformly in  $n$ . The corresponding Hamiltonians converge uniformly on  $K_M$ .*

*Proof.* Bernstein approximation gives uniform convergence of derivatives on the finite coordinate box. The entire cutoff prevents growth outside a slightly larger box while leaving  $K_M$  unchanged. The coordinate tests are chosen from the bounded collision-increment core, so a uniform derivative bound gives a uniform bound on every  $\Delta p + \psi$ . Dominated convergence applies to the exponential collision Hamiltonian.  $\square$

**1.4. Generator convergence.** Let the kinetic Hamiltonian be

$$\mathcal{H}(f; p, \psi) = \langle p, -v \cdot \nabla_x f \rangle + \int (e^{\Delta p + \psi} - 1) dA_f.$$

**Theorem 1.4** (Perturbed-test generator convergence). *For every analytic core test there is one exact forward-corrected sequence satisfying both relaxed upper and lower generator inequalities. By Lemma 1.3 the result extends to finite-coordinate  $C_b^3$  tests with bounded collision increments. The convergence is local in the weak-energy topology and uniform on action compact sets.*

*Proof.* Theorem 1.2 gives equality up to  $o(1)$  for analytic core tests, hence both relaxed inequalities. Approximate a nonanalytic test on the compact containment set by Lemma 1.3, first pass to the microscopic limit for fixed approximation, and then send the approximation index to infinity. Uniform bounded collision increments control the exponential nonlinearity.  $\square$

**1.5. Action compactness.** Let

$$\mathcal{I}(f, \Gamma) = I_0(f_0) + \int_0^T \int \ell(d\Gamma/dA_f) dA_f dt.$$

The topology is ordinary weak convergence of time marginals and contact measures; energy is a lower-semicontinuous containment function, not part of the convergence definition.

**Lemma 1.5** (Compact action sublevels). *Sets with bounded mass, energy, and  $\mathcal{I}$  are compact in the declared path topology. The balance equation is closed there, and concatenation at a common marginal preserves the action additively.*

*Proof.* Energy gives velocity tightness at every time. The entropy inequality with a bounded indicator source bounds contact mass on short time intervals by  $C|I| + o(1)$  uniformly on an action sublevel, because  $A_f(I)$  is controlled by mass and energy. The balance equation then gives equicontinuity for a countable convergence-determining family of bounded tests. Prokhorov and Arzela–Ascoli give compactness. Lower semicontinuity of perspective entropy and weak continuity of  $A_f$  on energy-tight sets close the balance relation. Concatenation adds the integrals.  $\square$

**1.6. Bounded-gradient comparison.** Choose bounded core coordinates  $\phi_j$  and define

$$d_B(f, g) = \sum_{j \geq 1} 2^{-j} \vartheta(\langle f - g, \phi_j \rangle), \quad \vartheta(r) = \sqrt{1 + r^2} - 1.$$

For  $\eta > 0$  use the Tataru-type penalty

$$\Psi_\eta(f, g) = \eta^{-1} \omega(\eta d_B(f, g)),$$

where  $\omega'(r) \leq 1$  and  $\omega(r) = r$  near zero. Its derivative coefficients are bounded by  $2^{-j}$  independently of  $\eta$ ; the number of active coordinates grows, but no coefficient blows up.

**Theorem 1.6** (Comparison in the bounded Hamiltonian core). *Bounded upper-semicontinuous subsolutions and bounded lower-semicontinuous supersolutions of the kinetic Hamilton–Jacobi equation satisfy comparison on every energy containment set. Every doubled-variable cotangent generated by  $\Psi_\eta$  has a collision increment bounded uniformly in  $\eta$ .*

*Proof.* At a doubled maximum, differentiate the bounded-gradient penalty. Since  $|\vartheta'| \leq 1$  and  $|\omega'| \leq 1$ , the source coefficient of  $\phi_j$  is at most  $2^{-j}$ . Summability and the bounded increments of  $\phi_j$  give a uniform bound for  $\Delta p$ . The Hamiltonian continuity lemma therefore applies without an  $\eta^{-1/2}$  escape. Send the finite-coordinate cutoff to infinity, then the containment penalty to zero, and finally  $\eta$  to zero. The standard doubling argument yields the contradiction to a positive maximum.  $\square$

**1.7. The nonlinear semigroup and microscopic convergence.** Define the Lax–Oleinik value

$$(V_t F)(f) = \sup_{(g, \Gamma): g_0 = f} \{F(g_t) - \mathcal{I}_{[0, t]}(g, \Gamma)\}.$$

**Theorem 1.7** (Kinetic semigroup limit). *The action value is attained, satisfies the semigroup law, and is the unique viscosity solution generated by  $\mathcal{H}$ . The microscopic augmented-law log-Laplace semigroups converge locally uniformly on compact containment sets to  $V_t$ . The lower recovery uses the upstream B2 exact projected tilt and balance-preserving regularization, not any B4 conclusion.*

*Proof.* Lemma 1.5 gives attainment and compact concatenation. Dynamic programming gives the semigroup law. Comparison from Theorem 1.6 gives uniqueness. The nonlinear-generator convergence of Theorem 1.4, compact containment, and the standard half-relaxed-limit theorem give microscopic convergence. For the lower inequality use the B2 normalized finite-dimensional tilt followed by its exact feasible regularization; this input is already proved upstream.  $\square$

**Theorem 1.8** (Round-seven B4 closure). *The exact state contains the contact ledger; the hierarchy corrector solves a forward terminal-value equation*

*with zero terminal remainder; the infinite series converges in graph norm; nonanalytic cylinders are approximated on action compacta; doubled cotangents stay in a uniformly bounded exponential core; and the nonlinear kinetic semigroup follows without a negative-time hierarchy or circular recovery argument.*

*Proof.* Combine Lemmas 1.1, 1.3, and 1.5 with Theorems 1.2, 1.4, 1.6, and 1.7.  $\square$

#### SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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